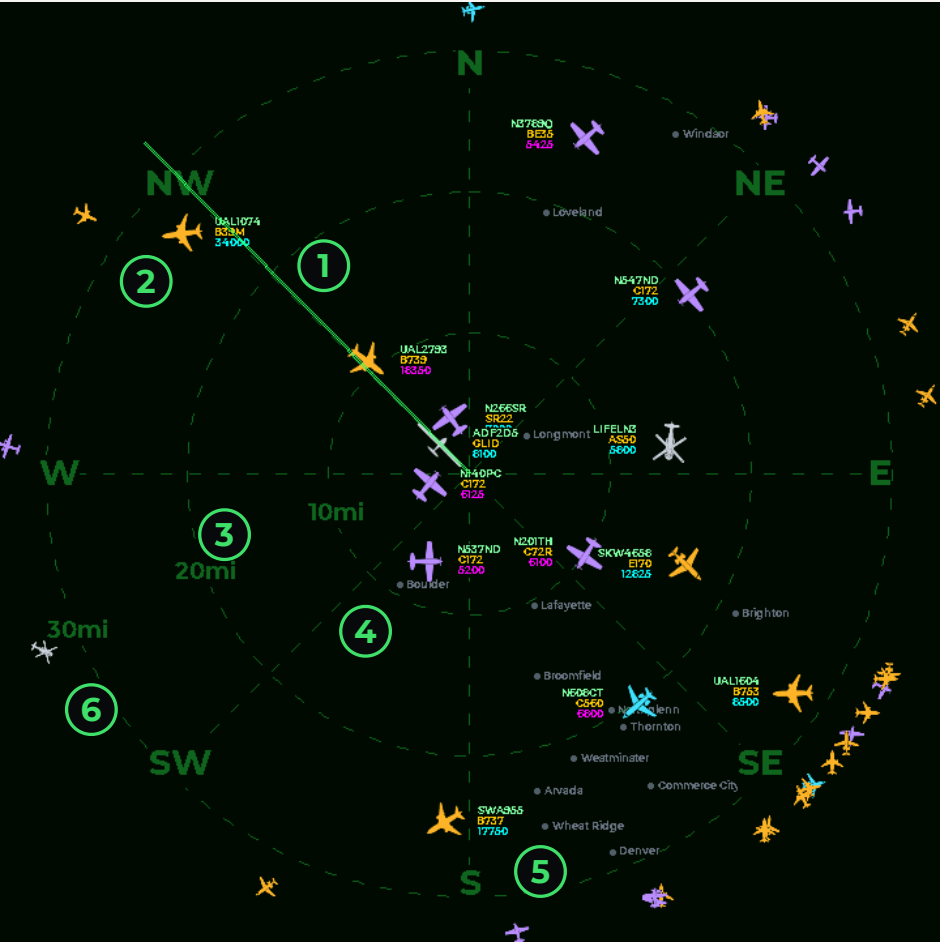




An actual capture from the device — 720 × 720 pixels of real traffic at 30 mile range. The centre of the circle is this building.

WHAT THE COLOURS MEAN

- Airlines & freighters
- Private & business jets
- Propeller & turboprop
- Helicopters, military, gliders

Military and emergency traffic overrides these colours and **pulses orange or red** instead.

THE SHORT VERSION

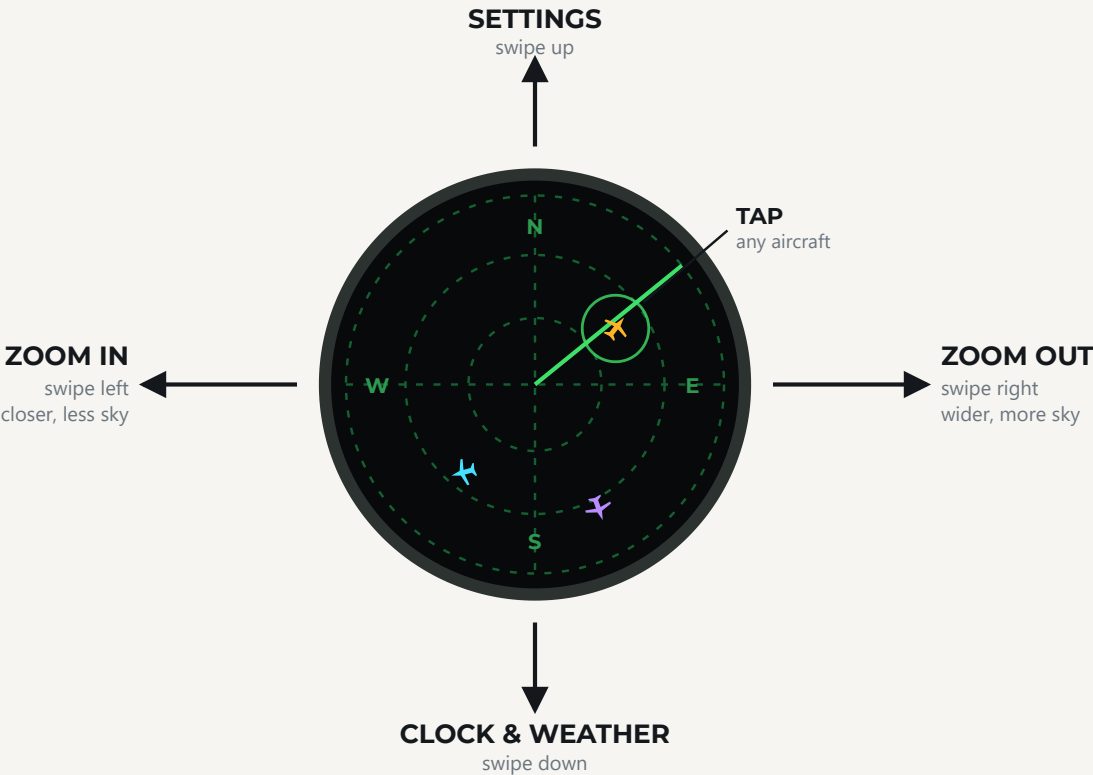
Aircraft constantly broadcast who and where they are. This round screen listens for those broadcasts, works out where each airplane sits relative to **this office**, and paints it on a radar scope. **Nothing here is simulated or replayed** — if a plane is on the glass, it is overhead.

THE SIX THINGS ON SCREEN

- 1 The sweep arm**
Rotates once every six seconds. Each aircraft only moves when the beam passes it, then holds still until the next lap — exactly how a real radar scope behaves.
- 2 An aircraft, and its tag**
The silhouette matches the real aircraft type and is turned to its actual heading. Beside it: callsign, type code and altitude in feet — **cyan** if level or climbing, **magenta** if descending.
- 3 Range rings**
Distance from this building, in miles. Currently 10 / 20 / 30 — swipe to change.
- 4 Towns underneath**
A dim map of nearby places, so the picture lines up with the geography you know.
- 5 Compass**
North is up. The scope is oriented to true north, not to the wall it hangs on.
- 6 Marks on the rim**
Traffic further out than the outer ring, parked at the edge on its correct bearing. Zoom out and it comes onto the scope properly.

How to drive it

Touch only — there are no buttons and no knob. Six gestures do everything.



TAP AN AIRCRAFT

Who is that?

Opens the flight-detail card: callsign, airline name and logo, where it came from and where it is going, aircraft type, altitude and speed. Swipe down or right to go back.

PRESS & HOLD ~3½ SECONDS

Only if the Wi-Fi changes

Puts the device into Wi-Fi setup so it can be pointed at a new network. Harmless by accident — it gives up after five minutes and reconnects on its own.

LAST RESORT

Blank, frozen, or plainly sulking?

Unplug the USB-C cable, count to five, plug it back in. It finds the Wi-Fi and rebuilds the picture within a minute or so, and no settings are lost by restarting it.

THE FIVE SCREENS

Radar — home. Everything comes back here on its own.

Clock & weather — time, date, current conditions, sunrise and sunset.

Forecast — three days out. From the clock, swipe right.

Flight detail — one aircraft, in full. Tap a blip.

Settings & about — range, units, colours, brightness. Swipe up.

HOW FAR IT CAN SEE

Range	Rings	Good for
5 mi	2 / 3 / 5	Only what is nearly overhead
10 mi	3 / 7 / 10	Local circuits and approaches
20 mi	7 / 13 / 20	A comfortable everyday picture
30 mi	10 / 20 / 30	The whole metro — current setting

Swipe left to come in, right to go out. It stops at both ends rather than wrapping around.

Three things that look like faults but aren't

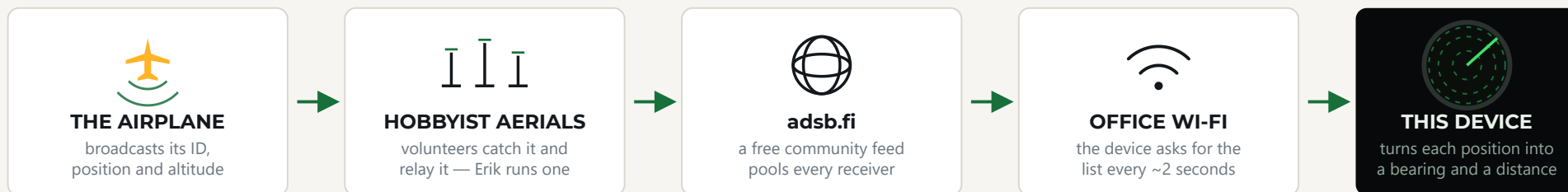
It went to the clock	The sky is empty. It returns to the radar the moment traffic shows up.
It left my screen	Every screen drifts back to the radar after about ten seconds untouched.
A plane sat still	It is waiting for the sweep arm. Give it up to six seconds.

How it **actually** works

From an airplane's radio transmitter to this desk, in a couple of seconds.

HOW IT WORKS

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NEW DATA every ~2 s

SWEEP one lap per 6 s

SCREEN ~30 frames per second

CAPACITY 64 aircraft, nearest first

LAG a few seconds

Why does it sweep like that?

A real radar has a physical dish that can only look one way at a time, so its display refreshes a target only when the beam comes round to it. This device already knows where everything is — it withholds the update deliberately, and repaints each aircraft as the arm crosses its bearing. That is the whole illusion, and it is why a blip can sit motionless for a moment and then jump.

The honest limitation: this is not really a radar. It cannot see anything that isn't voluntarily broadcasting — notably some military aircraft, and light aircraft old enough to carry no transponder.

What's inside

Brain	Espressif ESP32-P4, dual-core RISC-V
Memory	32 MB flash · 32 MB PSRAM
Screen	3.4" round, 720 × 720, capacitive touch
Radio	ESP32-C6 Wi-Fi 6 co-processor
Power	USB-C — the only cable it has
Software	Purpose-built firmware, no operating system

It carries its own aircraft-type database, airport list and town map in flash, so the silhouettes and place names need no internet at all. Only live traffic, weather and route lookups go out to the network.

Adjusting it

Everything on the device can also be set from a browser on the same Wi-Fi at <http://flightscnr.local/> — range, units, accent colour, brightness, night dimming, a watch list for particular callsigns, and the alert rules. Changes apply straight away, without a reboot.

A quieter option lives in there too: it can be pointed at a local ADS-B receiver on the LAN instead of the internet feed, in which case an aerial on the roof is the only source and nothing leaves the building.

Is it watching us?

No. It has no camera and no microphone, and it only ever reads a public list of aircraft positions. The one thing it knows about this room is the coordinates someone typed in.

Why is that plane so low?

Altitude is above sea level, not above the car park. Around here the ground is already about 5,000 ft up, so a light aircraft in the pattern reads 6,000-something.

Please leave it plugged in

It is not a phone charger. Pulling the cable costs it a minute of finding the network and the sky again — and the room loses its favourite distraction.